

4 Extended Link Services

4.1 Introduction

An Extended Link Service (ELS) request solicits a destination Nx_Port to perform a function. An ELS reply shall be transmitted in response to an ELS request, unless otherwise specified. Each request or reply is composed of a single Sequence with the ELS_Command code being specified in the first word of the Payload of the first frame of the Sequence. If Zoning is active in the Fabric (see FC-GS-5), an ELS response from a well-known address (e.g., the Domain Controller) shall only include data relating to Nx_Ports that are in the same zone(s) as the requesting Nx_Port.

Each Sequence may be composed of one or more frames. Normal rules for Exchange and Sequence management apply to ELS frames, Sequences, and Exchanges. An Accept (LS_ACC) to an ELS shall terminate the Exchange by setting the Last Sequence bit to one on the last frame of the reply. An ELS request and the corresponding reply shall be performed within a single Exchange. Normal rules for Exchange and Sequence management as defined in FC-FS-2 shall apply.

The TYPE field for ELS frames shall be set to 01h.

The R_CTL shall be set as specified in table 2.

Table 2 – Extended Link Services Routing Bits and Information Categories

R_CTL		Description
ROUTING	INFORMATION	
0010b	0001b	Solicited Data ^a
	0010b	Request
	0011b	Reply
	Others	Reserved
^a This value is only used by the Clock Synchronization Update (CSU) ELS.		

The first byte of the Payload (ELS_Command code) of the request or reply Sequence shall be as shown in table 3. The remainder of the Payload is ELS unique. Subsequent frames, if any, for a request or reply Sequence shall only contain additional Payload in the Payload field (i.e., the ELS_Command code is not repeated in each frame).

4.2 Extended Link Service requests

4.2.1 Introduction

A Sequence Initiator shall transmit an ELS Sequence in order to solicit the destination Nx_Port to perform a link-level function or service. Unless otherwise noted, Extended Link Service requests shall not be issued prior to completion of N_Port Login. Table 3 applies to ELSs sent to or received by all valid addresses, including well known addresses. **FLOGI is required before all ELSs if a Fabric is present.**

The LFA is used as the destination ID (D_ID) in the LINIT and LSTS ELS Request Sequences, and is used as the source ID (S_ID) in the Reply Sequences. No other Sequences shall be directed to a LFA.

Table 3 – ELS_Command codes

Value (Bits 31-24)	Description	Abbr.	Reference	N_Port Login Required
01h	Link Service Reject	LS_RJT	4.3.4	N/A
02h	Link Service Accept	LS_ACC	4.3.2	N/A
03h	N_Port Login	PLOGI	4.2.7	No
04h	F_Port Login	FLOGI	4.2.7	No
05h	Logout	LOGO	4.2.8	No
06h	Abort Exchange - obsolete	ABTX	N/A	N/A
07h	Read Connection Status	RCS	4.2.9	Yes
08h	Read Exchange Status Block - obsolete	RES	N/A	N/A
09h	Read Sequence Status Block - obsolete	RSS	N/A	N/A
0Ah	Request Sequence Initiative	RSI	4.2.13	Yes
0Bh	Establish Streaming	ESTS	4.2.6	Yes
0Ch	Estimate Credit	ESTC	4.2.5	Yes
0Dh	Advise Credit	ADVC	4.2.3	Yes
0Eh	Read Timeout Value	RTV	4.2.11	Yes
0Fh	Read Link Error Status Block	RLS	4.2.10	Yes
10h	Echo	ECHO	4.2.4	No
11h	Test	TEST	4.2.14	Yes
12h	Reinstate Recovery Qualifier	RRQ	4.2.12	Yes
13h	Read Exchange Concise	REC	4.2.42	Yes
14h	Reserved for legacy implementations ^a			
20h	Process Login	PRLI	4.2.20	Yes
^a Some early implementations of FCP-2 may have used the value 14h for SRR (Sequence Retransmission Request). This code is permanently reserved in this standard to avoid conflicts with such implementations. See FCP-3 for the standard implementation of SRR as an FC-4 Link Service.				

Table 3 – ELS_Command codes (Continued)

Value (Bits 31-24)	Description	Abbr.	Reference	N_Port Login Required
21h	Process Logout	PRLO	4.2.21	Yes
22h	State Change Notification - obsolete	SCN	N/A	N/A
23h	Test Process Login State	TPLS	4.2.22	Yes
24h	Third Party Process Logout	TPRLO	4.2.34	Yes
25h	Login Control List Management - obsolete	LCLM	N/A	N/A
30h	Get Alias_ID	GAID	4.2.26	No
31h	Fabric Activate Alias_ID	FACT	4.2.27	No
32h	Fabric Deactivate Alias_ID	FDACT	4.2.28	No
33h	N_Port Activate Alias_ID	NACT	4.2.29	No
34h	N_Port Deactivate Alias_ID	NDACT	4.2.30	No
40h	Quality of Service Request - obsolete	QoSR	N/A	N/A
41h	Read Virtual Circuit Status - obsolete	RVCS	N/A	N/A
50h	Discover N_Port Service Parameters	PDISC	4.2.31	Yes
51h	Discover F_Port Service Parameters	FDISC	4.2.32	Yes
52h	Discover Address	ADISC	4.2.33	Yes
53h	Report node Capability - obsolete	RNC	N/A	N/A
54h	Fibre Channel Address Resolution Protocol Request - obsolete	FARP_REQ	N/A	N/A
55h	Fibre Channel Address Resolution Protocol Reply - obsolete	FARP_REPLY	N/A	N/A
56h	Read Port Status Block - obsolete	RPS	N/A	N/A
57h	Read Port List - obsolete	RPL	N/A	N/A
58h	Report Port Buffer Condition	RPBC	4.2.37	Yes
^a Some early implementations of FCP-2 may have used the value 14h for SRR (Sequence Retransmission Request). This code is permanently reserved in this standard to avoid conflicts with such implementations. See FCP-3 for the standard implementation of SRR as an FC-4 Link Service.				

Table 3 – ELS_Command codes (Continued)

Value (Bits 31-24)	Description	Abbr.	Reference	N_Port Login Required
60h	Fabric Address Notification	FAN	4.2.15	No
61h	Registered State Change Notification	RSCN	4.2.18	No
62h	State Change Registration	SCR	4.2.19	No
63h	Report node FC-4 Types	RNFT	4.2.38	Yes
68h	Clock Synchronization Request	CSR	4.2.35	No
69h	Clock Synchronization Update	CSU	4.2.36	No
70h	Loop Initialize	LINIT	4.2.16	No
71h	Loop Port Control - obsolete	LPC	N/A	No
72h	Loop Status	LSTS	4.2.17	No
77h	Vendor Specific			N/A
78h	Request node Identification Data	RNID	4.2.23	No
79h	Registered Link Incident Report	RLIR	4.2.24	Yes
7Ah	Link Incident Record Registration	LIRR	4.2.25	Yes
7Bh	Scan Remote Loop	SRL	4.2.39	Yes
7Ch	Set Bit-error Reporting Parameters	SBRP	4.2.40	Yes
7Dh	Report Port Speed Capabilities	RPSC	4.2.41	Yes
7Eh	Query Security Attributes	QSA	see FC-SP	see FC-SP
7Fh	Exchange Virtual Fabrics Parameters	EVFP	4.2.43	N/A
80h	Link Keep Alive	LKA	4.2.44	No
^a Some early implementations of FCP-2 may have used the value 14h for SRR (Sequence Retransmission Request). This code is permanently reserved in this standard to avoid conflicts with such implementations. See FCP-3 for the standard implementation of SRR as an FC-4 Link Service.				

Table 3 – ELS_Command codes (Continued)

Value (Bits 31-24)	Description	Abbr.	Reference	N_Port Login Required
90h	Authentication ELS	AUTH_ELS	see FC-SP	see FC-SP
97h	Request Fabric Change Notification	RFCN	see FC-SP	see FC-SP
A0h	Define FFI Domain Topology Map	FFI_DTM	4.2.45	Yes
A1h	Request FFI Domain Topology Map	FFI_RTM	4.2.46	Yes
A2h	FFI AE Principal Switch Selector	FFI_PSS	4.2.47	Yes
A3h	FFI Map Update Registration	FFI_MUR	4.2.48	Yes
A4h	FFI Registered Map Update Notification	FFI_RMUN	4.2.49	Yes
A5h	FFI Suspend Map Updates	FFI_SMU	4.2.50	Yes
A6h	FFI Resume Map Updates	FFI_RMU	4.2.51	Yes
Others	Reserved			
^a Some early implementations of FCP-2 may have used the value 14h for SRR (Sequence Retransmission Request). This code is permanently reserved in this standard to avoid conflicts with such implementations. See FCP-3 for the standard implementation of SRR as an FC-4 Link Service.				

An ELS Protocol is composed of an ELS request Sequence and, for some requests, an ELS Reply Sequence. The last Data frame of an ELS request Sequence with a Reply Sequence shall transfer the Sequence Initiative to the Recipient in order to allow the reply to be transmitted (see FC-FS-2). If an ELS request Sequence that has a Reply Sequence is transmitted without the transfer of Sequence Initiative, the Recipient shall abort the Exchange and not perform the request.

The following ELS requests and their replies shall be supported by an Nx_Port (all others are optional for purposes of this standard, other standards or Technical Reports may require further ELS support):

- a) FLOGI
- b) PLOGI
- c) LOGO

An Nx_Port receiving an ELS request shall respond to it in accord with table 4, depending on its N_Port Login state with the Nx_Port sending the ELS request and the PLOGI requirement for the ELS specified in table 3.

Table 4 – Responses to Received ELSs

N_Port Login Required? (see table 3)	Logged in with Source N_Port?	
	Yes	No
Yes	Respond as appropriate for the ELS and the current state of the Nx_Port	If a reply sequence is defined for the ELS, originate a LOGO ELS Exchange to the sender of the received ELS or reply with an LS_RJT ELS Sequence with a reason code of "Unable to perform command request" and a reason code explanation of "N_Port Login required". If a reply sequence is not defined for the ELS, it shall be discarded
No	Respond as appropriate for the ELS and the current state of the Nx_Port.	Respond as appropriate for the ELS and the current state of the Nx_Port.

An Nx_Port is not required to generate and send the PLOGI ELS request. However, if an Nx_Port receives a PLOGI ELS request, the Nx_Port shall respond with a LS_ACC Link Service Reply, or with LS_RJT. LS_RJT shall not be issued with a reason code of "Command not supported" in response to a PLOGI.

NOTE 1 – If an Nx_Port that does not generate PLOGI, is in a point-to-point topology, and has an N_Port_Name greater than the other Nx_Port's, the other Nx_Port may timeout, waiting to receive PLOGI.

There is one ELS request Sequence in which a collision is possible with the other Nx_Port involved in the same target Exchange. This request is Request Sequence Initiative (RSI) (e.g., Nx_Port (A) may transmit an RSI request to Nx_Port (B) at the same time that Nx_Port (B) transmits an RSI request to Nx_Port (A) for the same target Exchange).

If such an instance occurs, the Originator Nx_Port of the target Exchange shall reject the RSI request sequence with an LS_RJT with a reason code of "command already in progress". The Responder Nx_Port of the target Exchange shall honor and process the RSI request Sequence normally.

4.2.2 Sequence and Exchange management

Extended Link Service communication shall observe all relevant rules of FC-FS-2.

All ELS requests, excluding ESTS, ESTC, and ADVC, and the corresponding replies shall be performed within a single Exchange, intended exclusively for the purpose of ELS processing (see FC-FS-2 for the procedure using ESTS, ESTC, and ADVC). The Advise Credit request may also be performed in a separate Exchange. Most ELS protocols are performed as a two Sequence Exchange. Each of these two Sequence Exchanges consist of a request Sequence by the Originator, transfer of Sequence Initiative, and a reply Sequence from the Responder that terminates the Exchange by setting the Last_Sequence bit (bit 20) in F_CTL.

More than one frame may be used to form a request or reply Sequence.

The following rules regarding Sequence and Exchange management apply to ELSs in addition to the rules specified in FC-FS-2:

- a) Request and Reply Sequences may be sent in any class of service;

NOTE 2 – Not all of the classes make sense for all ELSs, but there is no restriction.

- b) Reply frames and Sequences shall be transmitted in the same class as the request;
- c) If Login has not been completed successfully, the default Login values shall be used;
- d) If Login has completed successfully, the Originator of the Exchange shall use the Discard multiple Sequences Error Policy for all ELS Exchanges (see FC-FS-2);
- e) The Originator of an ELS Exchange shall detect an Exchange error following Sequence Initiative transfer if the Reply Sequence is not received within a timeout interval of $2 \times R_A_TOV$;
- f) If the Exchange Originator of an ELS Exchange detects an Exchange error, it shall abort the Exchange using ABTS-LS and retry the protocol of the aborted Exchange with a different Exchange;
- g) If the Sequence Initiator aborts a Sequence using ABTS due to receiving an ACK with the Abort Sequence bits set to 01b, the Sequence Initiator shall retry the Sequence after the BA_ACC is received for the aborted Sequence one time only. If the retry fails, the ELS Exchange shall be aborted using ABTS-LS;
- h) If the Sequence Initiator attempts to abort a Sequence using ABTS and it detects an E_D_TOV waiting for the ACK frame in response to the ABTS, it shall abort the Exchange using ABTS-LS, if conditions permit, and retry the original ELS with a different Exchange; and
- i) If the Sequence Initiator attempts to abort a Sequence using ABTS-LS and it detects an E_D_TOV waiting for the ACK frame in response to the ABTS-LS, it may retry the original ELS with a different Exchange.

4.2.3 Advise Credit (ADVC)

4.2.3.1 Description

The ADVC ELS is used to advise the destination Nx_Port of the estimated end-to-end Credit that the source Nx_Port requests to be allocated. The ADVC ELS request shall be a separate Sequence. It may also be requested in a separate Exchange. See FC-FS-2 for the usage of this ELS. The ADVC request may also be used independently from the Estimate Credit procedure (see FC-FS-2).

4.2.3.2 Protocol

- a) Advise Credit Request Sequence
- b) LS_ACC or LS_RJT Reply Sequence

4.2.3.3 Request Sequence

Addressing: The S_ID field designates the source Nx_Port requesting Credit revision. The D_ID field designates the destination Nx_Port.

Payload: The format of the Payload is shown in table 5. The Payload shall contain the requested end-to-end Credit in the end-to-end Credit field of the appropriate Class Service Parameters (see 6.6.5) as indicated by the Class Validity bit. For each class in which a revised end-to-end Credit is requested, the Class Validity bit shall be set to one. The recipient shall ignore the other Service Parameter fields.

Table 5 – ADVC Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	ADVC (0Dh)	00h	00h	00h
1	MSB	Common Service Parameters (16 bytes)		
..				
4				LSB
5	MSB	N_Port_Name (8 bytes)		
6				LSB
7	MSB	Node_Name (8 bytes)		
8				LSB
9	MSB	Class 1 Service Parameters (16 bytes)		
..				
12				LSB
13	MSB	Class 2 Service Parameters (16 bytes)		
..				
16				LSB
17	MSB	Class 3 Service Parameters (16 bytes)		
..				
20				LSB
21	MSB	Reserved (16 bytes)		
..				
24				LSB
25	MSB	Vendor Version Level (16 bytes)		
..				
28				LSB

4.2.3.4 Reply Sequence

LS_RJT: LS_RJT signifies rejection of the ADVC command.

LS_ACC: LS_ACC signifies successful completion of the ADVC function and permanently replaces the end-to-end Credit in effect for the current N_Port Login.

The format of the LS_ACC Payload is shown in table 6. The Payload shall contain the revised end-to-end Credit allocated in the Credit field for the appropriate Class Service Parameters as indicated by the Class Validity bit. The revised end-to-end Credit shall replace the end-to-end Credit for the current Login for the Nx_Port transmitting the LS_ACC Sequence (see 6). For each class in which a revised end-to-end Credit is updated, the Class Validity bit shall be set to one. The recipient shall ignore the other Service Parameter fields. This revised end to-end Credit value is determined by the destination Nx_Port based on its buffering scheme, buffer management, buffer availability, and Nx_Port processing time (see FC-FS-2).

Table 6 – ADVC LS_ACC Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	02h	00h	00h	00h
1	MSB Common Service Parameters (16 bytes)			
..				
4				LSB
5	MSB N_Port_Name (8 bytes)			
6				LSB
7	MSB Node_Name (8 bytes)			
8				LSB
9	MSB Class 1 Service Parameters (16 bytes)			
..				
12				LSB
13	MSB Class 2 Service Parameters (16 bytes)			
..				
16				LSB
17	MSB Class 3 Service Parameters (16 bytes)			
..				
20				LSB
21	MSB Reserved (16 bytes)			
..				
24				LSB
25	MSB Vendor Version Level (16 bytes)			
..				
28				LSB

4.2.4 Echo (ECHO)

4.2.4.1 Description

The Echo ELS requests the Recipient to transmit the Payload contents that follows the ELS_Command back to the Initiator of the Echo command in the same order as received using the LS_ACC Reply Sequence. The Echo ELS Request provides a means to transmit a Data frame and have the Payload content returned for a simple loopback diagnostic function. The Echo command shall be transmitted as a one frame Sequence and the LS_ACC Reply Sequence is also a one frame Sequence.

4.2.4.2 Protocol

- a) Echo Request Sequence
- b) LS_ACC or LS_RJT Reply Sequence

4.2.4.3 Request Sequence

Addressing: The D_ID field designates the destination of the request while the S_ID field designates the source of the request.

Payload: The format of the Payload is shown in table 7. If a Login with the destination Nx_Port does not exist, the maximum size of the ECHO data field shall be the default Buffer-to-buffer Receive Data_Field Size - 4 (i.e., 124 bytes). If a Login with the destination Nx_Port exists, the ECHO data field size is limited by the smallest Receive Data_Field Size supported by the destination Nx_Port, the Fabric, and the source Nx_Port for the class of service being used - 4 (i.e., the ECHO data field in the LS_ACC frame shall be equal in size to the ECHO data field size in the ECHO Request Sequence).

Table 7 – ECHO Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	ECHO (10h)	00h	00h	00h
1	MSB	ECHO data (up to max frame length - 4, any byte boundary)		
..				
n				LSB

4.2.4.4 Reply Sequence

LS_RJT: LS_RJT signifies rejection of the ECHO command.

LS_ACC: LS_ACC signifies successful completion of the ECHO function. The format of the LS_ACC Payload is shown in table 8. The Payload shall contain the ECHO data contained in the Payload of the ECHO Request frame.

Table 8 – ECHO LS_ACC Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	02h	00h	00h	00h
1	MSB	ECHO data		
..		(up to max frame length - 4, any byte boundary)		
n		Excludes word 0 of ECHO Payload.		LSB

4.2.5 Estimate Credit (ESTC)

4.2.5.1 Description

The ESTC ELS is used to estimate the minimum Credit required to achieve the maximum bandwidth for a given distance between an Nx_Port pair.

The class of the SOF of the ESTC Request identifies the class for which Credit is being estimated. The destination Nx_Port shall acknowledge Data frames as specified by its Login parameters. See FC-FS-2 for the usage of this frame.

4.2.5.2 Protocol

- a) Estimate Credit Request Sequence
- b) No Reply Sequence

4.2.5.3 Request Sequence

Addressing: The S_ID field designates the source Nx_Port requesting the Credit estimate. The D_ID field designates the destination Nx_Port specified in the Establish Streaming frame.

Payload: The format of the Payload is shown in table 9. The first word of the Payload of the first frame of the Sequence shall contain the ELS_Command code. The content of the Payload after the ELS_Command and for subsequent frames shall be valid data bytes.

Table 9 – ESTC Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	ESTC (0Ch)	00h	00h	00h
1	MSB	Any data		
..		(see FC-FS-2)		
n				LSB

4.2.5.4 Reply Sequence

None.

4.2.6 Establish Streaming (ESTS)

4.2.6.1 Description

The ESTS ELS requests a temporary allocation of Credit known as Streaming Credit large enough to perform continuous streaming of Data frames. The SOF of the ESTS Request identifies the class for which Credit is being estimated. See FC-FS-2 for the usage of this frame.

4.2.6.2 Protocol

- a) Establish Streaming Request Sequence
- b) LS_ACC or LS_RJT Reply Sequence

4.2.6.3 Request Sequence

Addressing: The S_ID field designates the source Nx_Port requesting Streaming. The D_ID field designates the destination Nx_Port addressed.

Payload: The format of the Payload is shown in table 10.

Table 10 – ESTS Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	ESTS (0Bh)	00h	00h	00h

4.2.6.4 Reply Sequence

LS_RJT: LS_RJT signifies rejection of the ESTS command

LS_ACC: LS_ACC signifies successful completion of the ESTS function. The format of the LS_ACC Payload is shown in table 11. The Payload shall contain Streaming Credit allocated in the Nx_Port end-to-end Credit field of the appropriate Class Service Parameters (see 6.6.5). The Class Validity bit, when set to one, identifies the class that contains the Streaming Credit. The recipient shall ignore the other Service Parameter fields.

Table 11 – ESTS LS_ACC Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	02h	00h	00h	00h
1	MSB	Common Service Parameters (16 bytes)		
4				LSB
5	MSB	N_Port_Name (8 bytes)		
6				LSB
7	MSB	node Name (8 bytes)		
8				LSB
9	MSB	Class 1 Service Parameters (16 bytes)		
..				
12				LSB
13	MSB	Class 2 Service Parameters (16 bytes)		
..				
16				LSB
17	MSB	Class 3 Service Parameters (16 bytes)		
..				
20				LSB
21	MSB	Reserved (16 bytes)		
..				
24				LSB
25	MSB	Vendor Version Level (16 bytes)		
..				
28				LSB

4.2.7 Login (FLOGI/PLOGI)

4.2.7.1 Description

The FLOGI/PLOGI ELS shall transfer Service Parameters from the initiating Nx_Port to the FC_Port associated with the D_ID. The FLOGI frame provides the means by which an Nx_Port may request Login with the Fabric (see 6.2). The PLOGI frame provides the means by which an Nx_Port may request Login with another Nx_Port prior to other Data frame transfers (see 6.3).

In order to Login with the Fabric and determine the Fabric operating characteristics, an Nx_Port shall specify the D_ID as the well-known F_Port_ID (i.e., FFFFFFFEh).

In order to direct the **Login ELS frame** to a Fibre Channel Service, an Nx_Port shall specify the N_Port_ID for the Nx_Port providing the service or the appropriate well-known Address Identifier (see FC-FS-2).

When an Nx_Port receives a Login from an Nx_Port, all open Sequences with the Nx_Port performing Login shall be abnormally terminated.

4.2.7.2 Protocol

- a) FLOGI or PLOGI Request Sequence
- b) LS_ACC or LS_RJT Reply Sequence

4.2.7.3 Request Sequence

Addressing: The S_ID field designates the source Nx_Port requesting Login. If unidentified, as in Fabric Login, binary zeros are used. The D_ID field designates the destination Nx_Port or Fx_Port of the Login.

Payload: The format of the Payload is shown in table 149. The Service Parameters are defined in 6.6.

4.2.7.4 Reply Sequence

LS_RJT: LS_RJT signifies rejection of the FLOGI or PLOGI Request Sequence

LS_ACC: LS_ACC signifies successful completion of the FLOGI or PLOGI Request Sequence. The format of the LS_ACC Payload is shown in table 149. The Service Parameters are defined in 6.6.

4.2.8 Logout (LOGO)

4.2.8.1 Description

The LOGO ELS provides a method for explicitly removing service between two Nx_Port_IDs or between an N_Port_ID and a Fabric. Logout releases resources, identifiers, and relationships associated with maintaining Service between an Nx_Port_ID and a destination Nx_Port_ID or Fabric.

If the S_ID and D_ID of a LOGO ELS are both Address_Identifier of Nx_Ports, the LOGO requests removal of service between the N_Port_ID specified in the LOGO Payload and the N_Port_ID specified in the D_ID. The N_Port_ID in the LOGO Payload may differ from the S_ID. This allows an Nx_Port to Logout its old Identifier using a new Identifier after its N_Port_ID has changed. Both the source Nx_Port and the destination Nx_Port of the Logout Request Sequence shall abnormally terminate all open Exchanges (see 6) that used the N_Port_ID indicated in the Payload of the Logout Request Sequence.

If either the S_ID or D_ID of a LOGO ELS is the F_Port well-known address (FFFFFFEh), the LOGO ELS requests Fabric logout and release of the previously assigned N_Port_ID specified in the LOGO Payload. An Nx_Port that requests or accepts explicit logout from the Fabric shall implicitly log out the N_Port_ID indicated in the Payload of the Logout Request Sequence from all other Nx_Port_IDs and abnormally terminate all open Exchanges (see 6) that used the N_Port_ID indicated in the Payload of the Logout Request Sequence.

4.2.8.2 Protocol

- a) Logout Request Sequence
- b) LS_ACC or LS_RJT Reply Sequence

4.2.8.3 Request Sequence

Addressing: For an explicit Nx_Port Logout, the S_ID field designates the source Nx_Port_ID requesting Logout and the D_ID field designates the destination Nx_Port of the Logout Request. For an explicit Fabric Logout originated by an Nx_Port, the S_ID field shall be the Nx_Port_ID to be logged out (i.e., the same as the N_Port_ID in the Payload) and the D_ID shall be the F_Port Well-known address (i.e., FFFFFFFEh). For an explicit Fabric Logout originated by the Fabric, the S_ID field shall be the F_Port Well-known address (i.e., FFFFFFFEh) and the D_ID shall be the Nx_Port_ID to be logged out (i.e., the same as the N_Port_ID in the Payload).

Payload: The format of the Payload is shown in table 12.

Table 12 – LOGO Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	LOGO (05h)	00h	00h	00h
1	Reserved	N_Port_ID		
2	MSB	N_Port_Name		
3		(8 bytes)		
				LSB

4.2.8.4 Reply Sequence

LS_RJT: LS_RJT signifies rejection of the LOGO command.

LS_ACC: LS_ACC signifies that service has been removed for the N_Port_ID indicated in the payload of the LOGO ELS. When the LOGO ELS requested Fabric Logout, LS_ACC signifies that the N_Port_ID specified in the Payload of the LOGO ELS has been logged out from the Fabric and released. The format of the LS_ACC Payload is shown in table 13.

Table 13 – LOGO LS_ACC Payload

Bits Word	31 .. 24	23 .. 16	15 .. 08	07 .. 00
0	02h	00h	00h	00h

4.2.9 Read Connection Status (RCS)

4.2.9.1 Description

The RCS ELS requests the Fabric Controller to return the current dedicated connection status for the Nx_Port specified in the Payload of the RCS frame. The RCS Request provides the means by which an Nx_Port may interrogate the Fabric for the Connection status of other Nx_Ports within the Fabric.